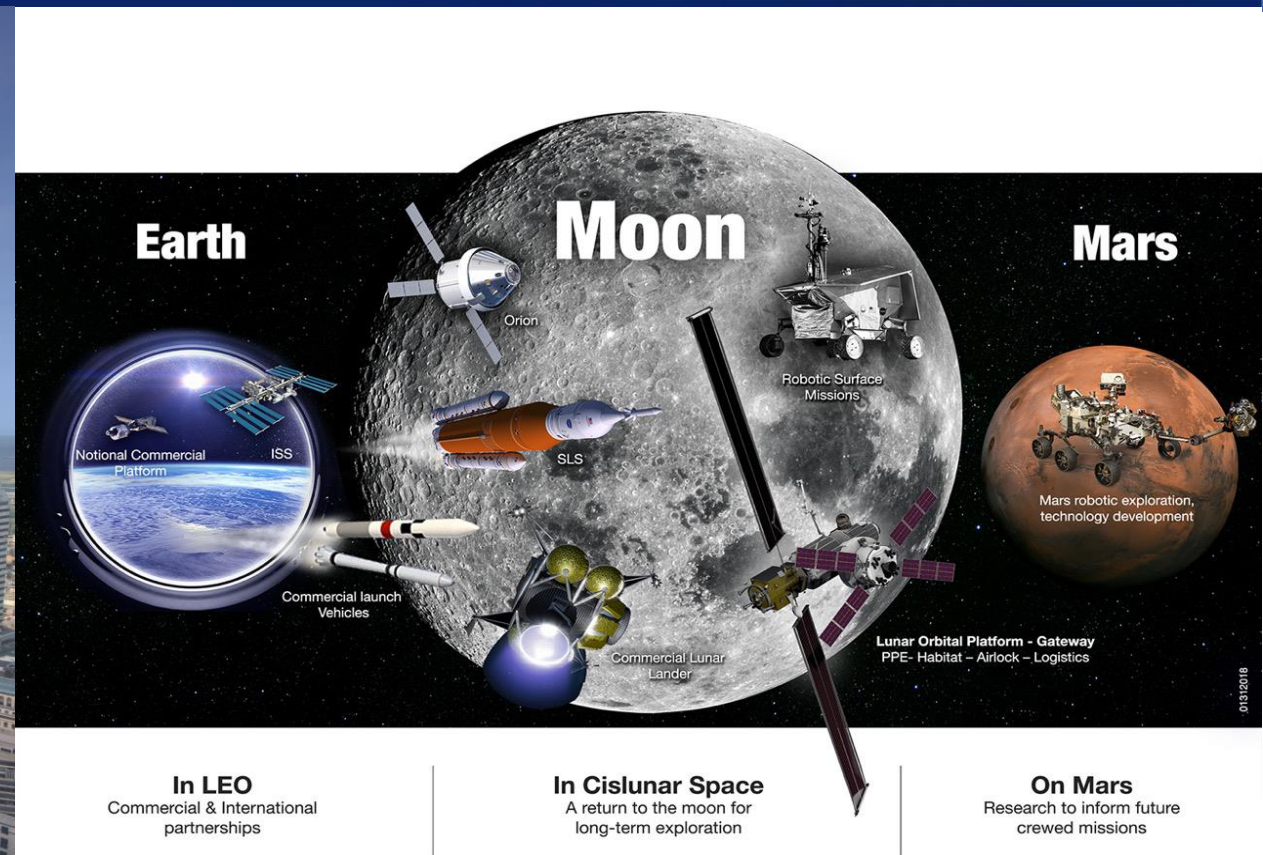


Energy Storage for NASA Missions



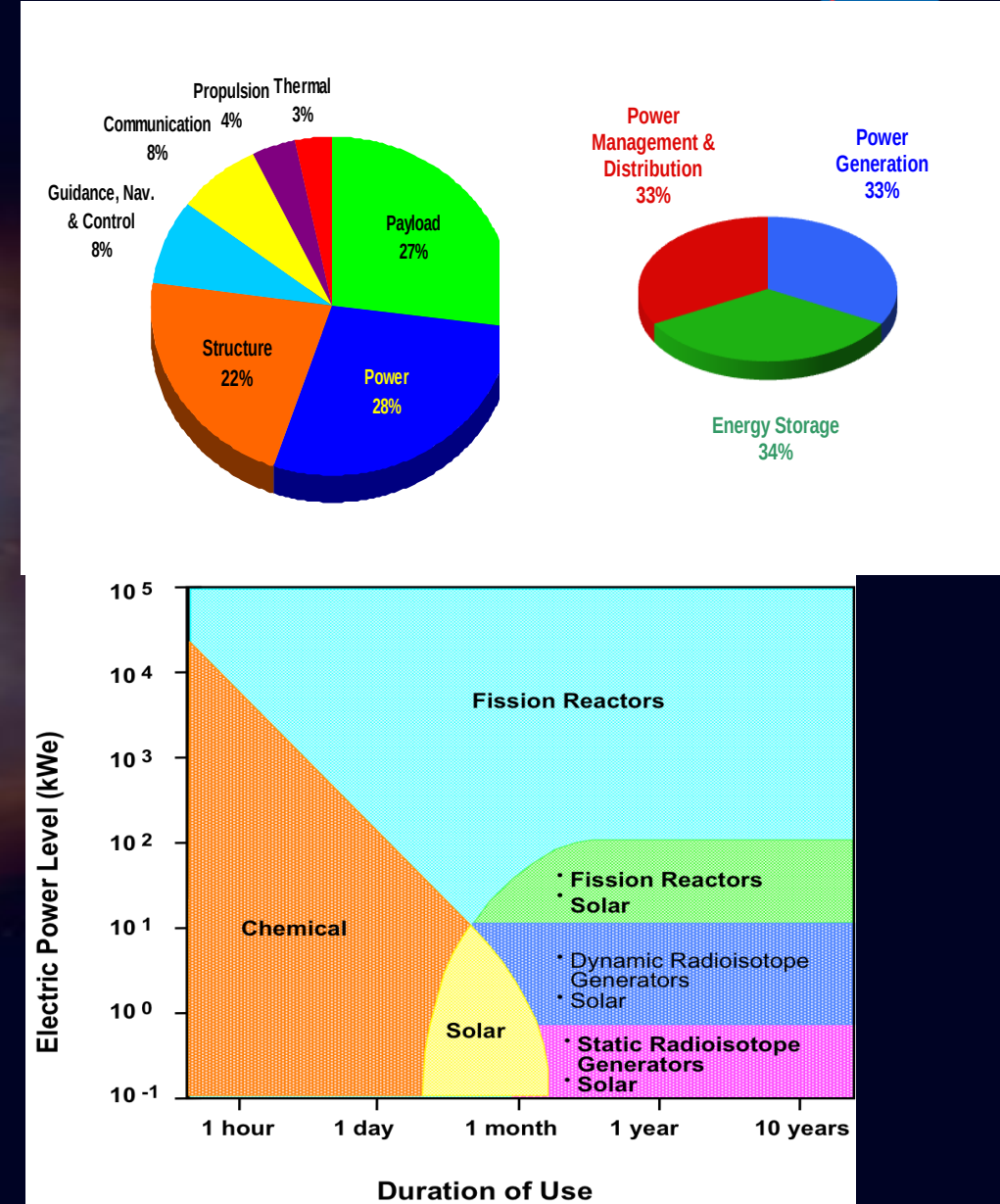
Dr. Vadim F. Lvovich
NASA Glenn Research Center

ARPA-E IONICS Annual Program Review, November 9-10, 2020

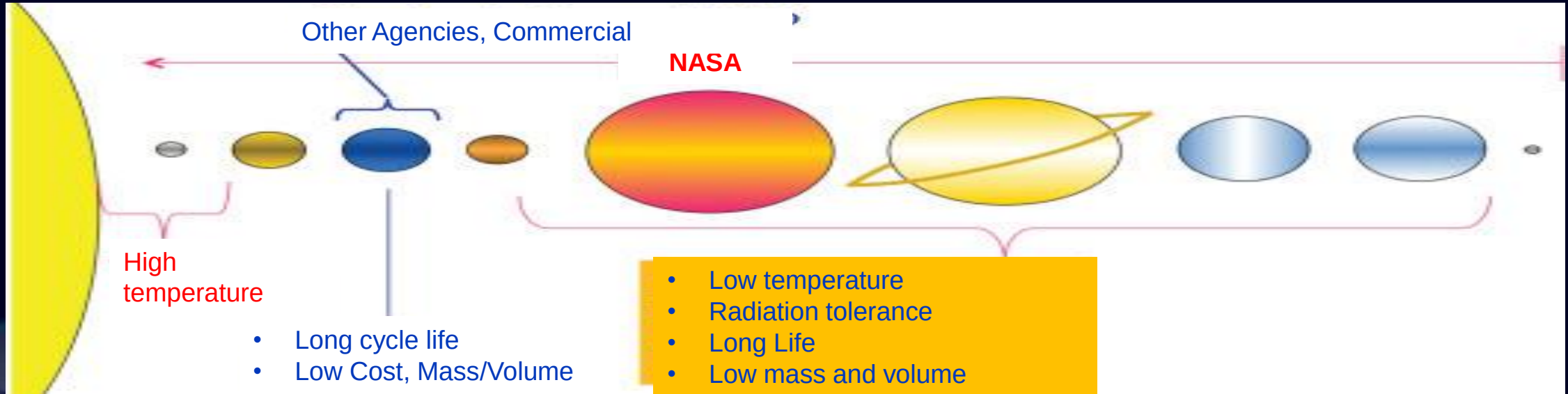
Space Power Technology Area Overview



- Power system is 20-30% of spacecraft mass & cost
- The major power subsystems are:
 - Power Generation/Conversion
 - Energy Storage
 - Power Management and Distribution
- Space missions need a variety of power solutions
 - Solar power systems
 - Nuclear power systems
 - Batteries
 - primary and secondary
 - Fuel Cells
 - New Technologies



NASA Goals in Batteries and Energy Storage



NASA's energy storage needs span a greater range of environments and cycle requirements than other organization's applications.

Energy storage technologies are core to every aerospace mission, and their mass is often referred as a barrier to achieving mass efficient systems

Several key NASA applications require very high specific energy (>500 Wh/kg) with enhanced safety, while commercial HEV-driven market requires low cost, long cycle life, with specific energy ~ 250 Wh/kg.

NASA Demands Very High Specific Energy Batteries and Fuel Cells



Electric Aviation 500 – 750 Wh/kg

- Green aviation – Less noise, lower emissions, high efficiency
- Hybrid / All-electric aircraft – Limited by mass of energy storage system
- Commercial aviation – Safe, reliable, lightweight on-board electric auxiliary power unit



Extravehicular Activities (Spacesuit power) >400 Wh/kg

Required to enable untethered EVA missions lasting 8 hours within strict mass and volume limitations.

Astronaut life support
Safety and reliability are critical
100 cycles



Landers and Rovers, Robotic missions, In- space habitats >500 Wh/kg

Batteries are expected to provide sufficient power for life support and communications systems, and tools including video and lighting

- >100 cycles

NASA future mission requirements far exceed the capabilities of lithium-ion chemistries

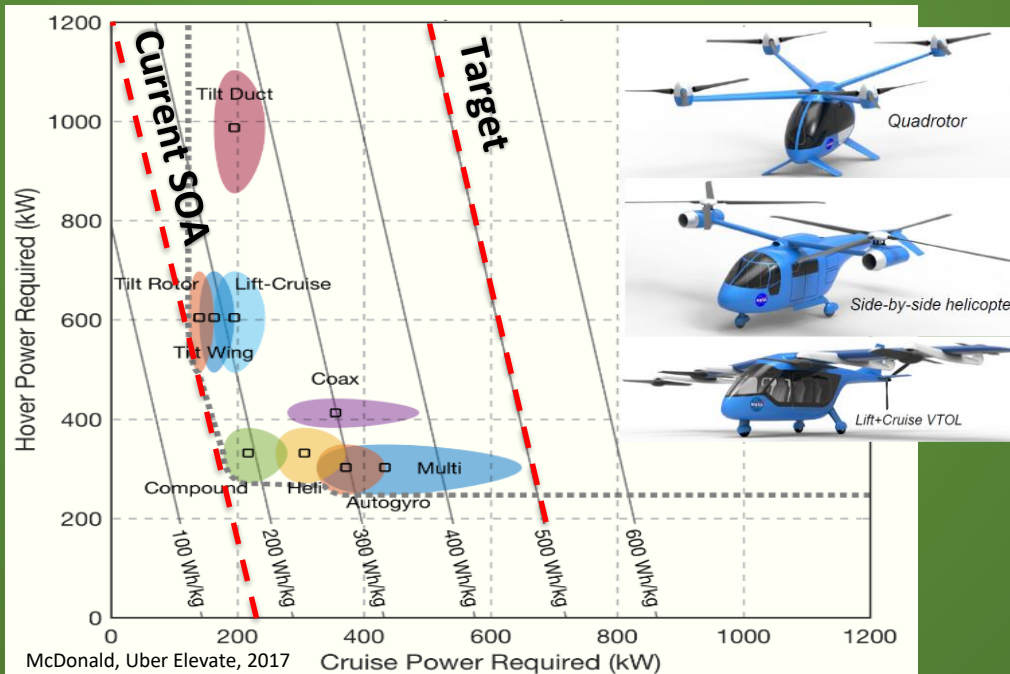
- Progress in these areas requires advances in safe, very high energy batteries and fuel cells

The Problem

Current SOA batteries are not designed to meet the unique NASA performance & safety requirements

Battery **Performance** Requirements

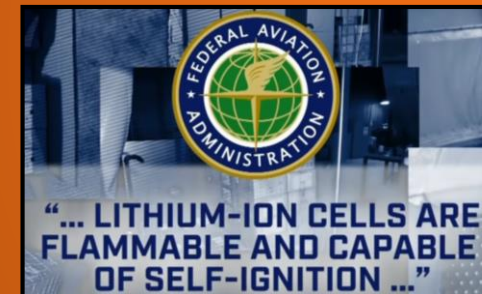
- ❑ NASA Battery Workshop 2017 and industry representatives state “The primary barrier to electric aviation is battery performance”
- ❑ SOA lithium ion batteries do not meet energy density requirements needed to enable electric aircraft designs
- ❑ Unique flight critical metrics (e.g. high power) required



Vehicle Performance & Efficiency

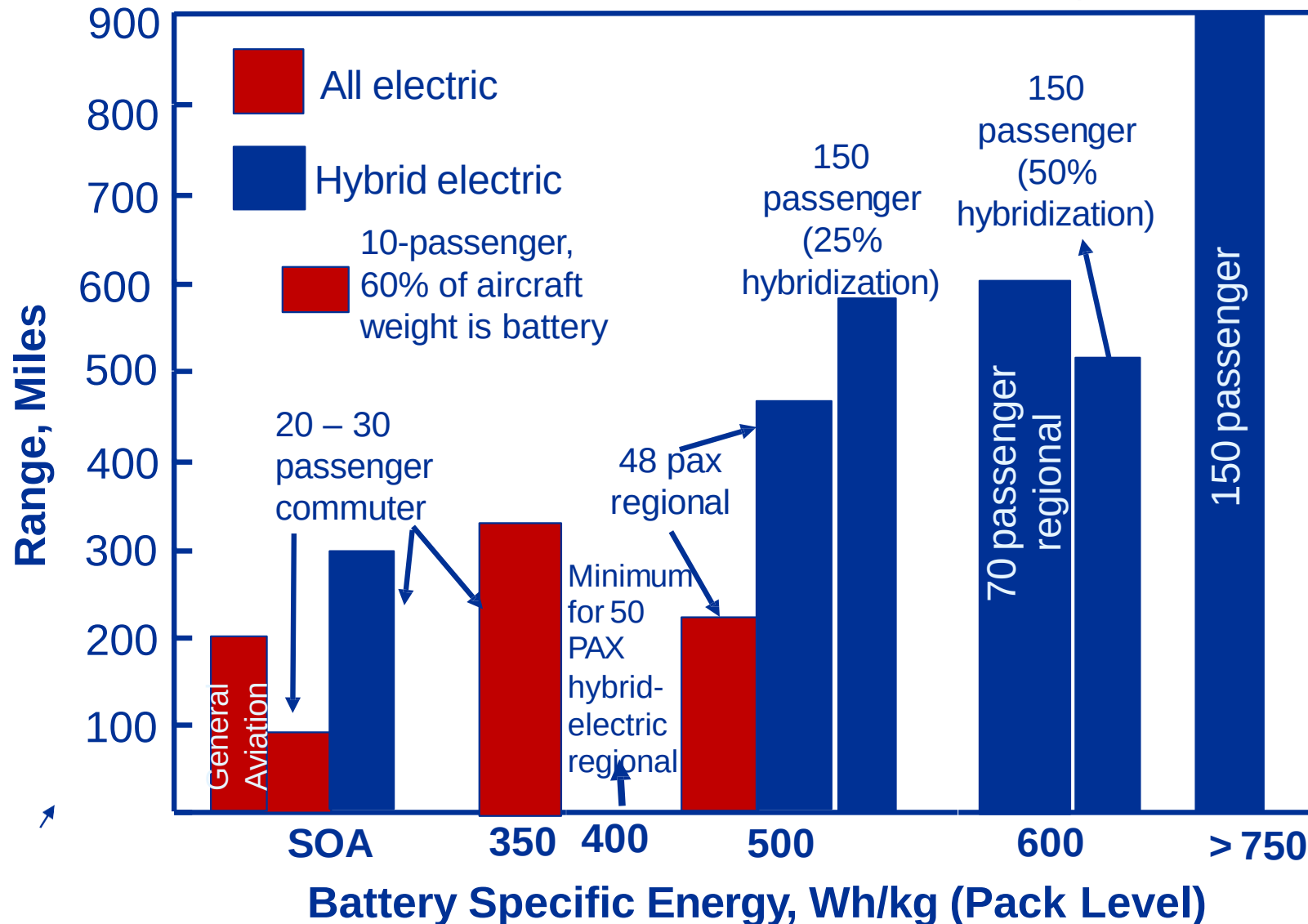
Battery **Safety** Requirements

- ❑ Current batteries under development will always have fire safety challenges due to flammable electrolytes used
- ❑ Safety is required for aerospace applications
- ❑ SOA lithium ion batteries have caused a number of safety incidents on aircraft
- ❑ Parasitic weight from excess packaging and cooling is undesirable



Specific Energy Requirement for Conventional Takeoff and Landing Electrified Aircraft

- Taken from different sources with different assumptions



- Battery requirement is a function of degree of hybridization
- Potential hybrid electric short flight (300 – 400 miles) opportunities for large regional and 150 Passenger single aisle with 400 – 500 Wh/kg specific energy
- For hybrid electric large aircraft with battery used during takeoff only, 350 – 400 Wh/kg at pack level required
- System analysis needed to identify the opportunity space for hybrid electric large regional and 150-passenger aircraft

Evolution of Electrified Aircraft Market with Improvements in Battery Technology

Battery Pack Specific Energy

SOA (150 – 170 Wh/kg)

Potential Missions



Potential Market Introduction

Initial commercial introduction possible for all-electric with limited range and payload, extended capability with hybrid-electric

300 Wh/kg



All-electric eVTOL urban air mobility with 4 passenger and 50+ mile range; 20-passenger all-electric commuter

400 Wh/kg

Sweet spot for eVTOL



Desired capability for all-electric eVTOL urban air mobility, long-range all-electric commuter, Initial version of small hybrid-electric regional

500 Wh/kg



Expansion to various classes of hybrid-electric regional aircraft, short-range 150 Passenger, single aisle hybrid-electric aircraft

> 700 Wh/kg

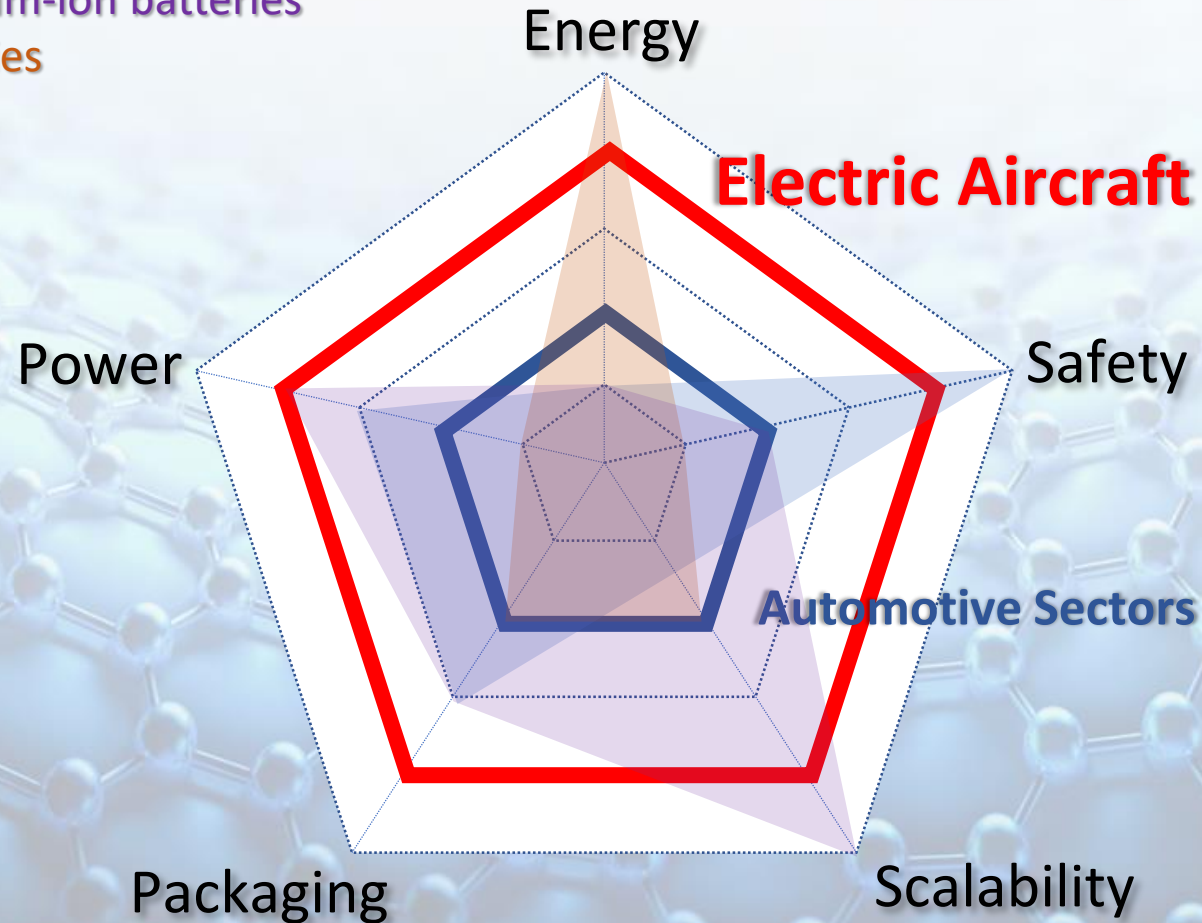


Single aisle, 150 passenger single-aisle aircraft, long range

The Big Question

How do we meet **ALL** demanding battery needs of electric aircraft?

- State-of-the-art lithium-ion batteries
- Lithium sulfur batteries
- Solid state batteries



Battery Needs for Electric Aviation (Beyond Current DOE and Industry Investment) – Workshop Output

Technology Needs	
System Level Modeling	Identify reference mission for each aircraft class and conduct system analysis to pinpoint opportunities as a function of battery capabilities
	Link aircraft propulsion models to battery performance and life models to better define interaction of batteries under aviation conditions
Next Gen Li – Performance and Safety	Evaluate high discharge operation of Li-ion cells, especially focused on eVTOL applications and develop possible solutions (e.g., new chemistries with high specific energy and power density capability or new thermal management solutions)
	Evaluate performance of next generation Li-ion (silicon, high Ni NMC) under aviation conditions and examine/model failure modes and safety aspects
Packaging and Integration	Develop efficient designs to package cells to modules and packs taking advantage of emerging approaches such as 3-D printing, bipolar designs and multifunctionality
	Augment R&D on all solid state batteries to examine new designs, manufacturing approaches, high temperature operations, etc.
Beyond Next Gen Li	Develop chemistries with 3-5X higher specific energy, including evaluating the possibility of enabling high energy primary batteries, sulfur-based chemistries, and hydrogen carriers with the aim of providing solutions for large regional and 737-class aircraft

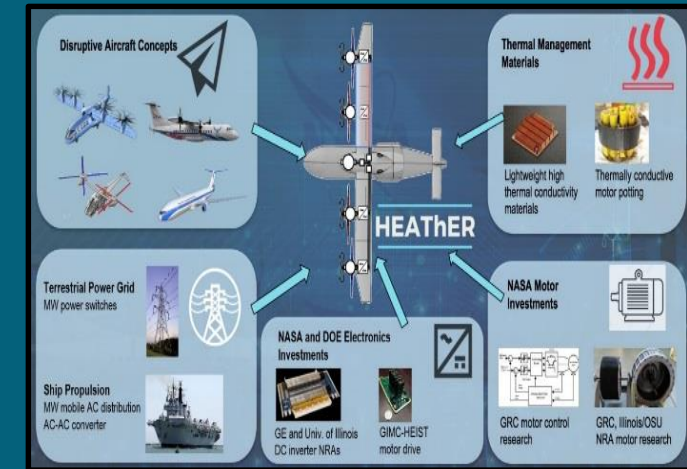
Transformational Aeronautics Concepts

Convergent Aeronautics Solutions Project

Fast moving 2 year activities that include multiple NASA centers, industry and academia focused on feasibility assessment

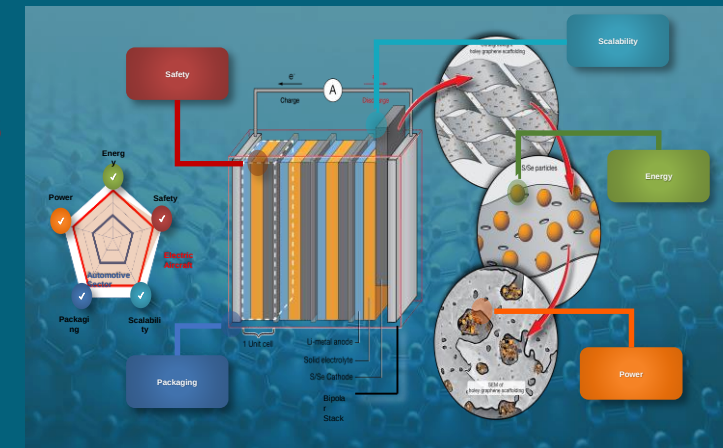
Current:

- **HEATHER**: High-efficiency Electrified Aircraft Thermal Research. Reduce power conversion requirements and use low-loss electrical components to enable local thermal management solution for MW-class EAP
- **SABERS**: Solid-State Architecture Batteries for Enhanced Rechargeability and Safety. Develop a solid-state bipolar battery stack based on novel Li-S/Se chemistry and a non-flammable electrolyte for UAM (FY20 new start)
- **SPARRCI**: Sensor-based Prognostics to Avoid Runaway Reactions & Catastrophic Ignition. Enable safe battery operation with higher specific energy via embedded sensors and machine learning (FY20 new start)



HEATHER

SABERS



Transformational Aeronautics Concepts

Completed Convergent Aeronautics Solutions Projects Activities:

- **AQUIFER:** Aqueous QUick-Charging Battery Integration For Electric Flight Research
- **LiON:** The Li-O₂ Batteries for NASA Electric Aircraft project targeted leveraging modern computational materials development methods, combined with experimental battery components design and evaluation tools to develop radically advanced compatible cathode and electrolyte materials, build cells and demonstrate the Li-Air battery packs.
- **FUELEAP:** Fostering Ultra Efficient, Low-Emitting Aviation Power combined the technical advancements in SOFC, high-yield fuel reformers, and hybrid-electric aircraft architectures to develop an integrated power system.
- **M-SHELLS:** Multifunctional Structures for High Energy Lightweight Load-bearing Storage evaluated the feasibility of a structural hybrid super-capacitor concept drawn from recent nano-technology advances in both electrochemistry and microstructures to create a transformative multi-functional airframe material which carries structural load and stores energy to save weight and enable electric aeropropulsion
- **CAMIEM:** Compact Additively Manufactured Innovative Electric Motor manufacturing offers the potential to radically change the motor designs so that they have compact designs, multi-material components, innovative cooling, and optimally designed and manufactured components. The goal was to assess the feasibility for utilizing additive manufacturing (AM) methods to enable new motor designs.
- **HVHEP:** activity demonstrated feasibility of variable frequency, variable high voltage power system using doubly fed electric machines (DFM), and provided 90% reduction in power electronic weight for 35 megawatt system. DFM enables multi-phase (9) ac operation, with enhanced control and integrated protection methods for improved safety through minimization of fault energy.



Camien Additively
Manufactured Motor

Concluding Thoughts

- Pack specific energy of 300 Wh/kg is achievable within reasonable timeframe
 - Will enable several classes of eVTOL for urban air mobility
 - Will enable several classes of 1-20 passenger and regional aircraft depending on degree of hybridization and range
- Pack specific energy of 400 Wh/kg will probably require maturation of all solid state technology or Li-S with liquid electrolyte
- No clear path for achieving pack specific energy greater than 500 Wh/kg
- Performance, safety, and life of developmental batteries under aircraft operating conditions need to be evaluated, particularly under high C-rating operation
- Opportunities for better packaging and integration